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Unemployment insurance and cigarette smoking

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ABSTRACT

We provide evidence for the causal relationship between unemployment insurance (UI) and individuals' smoking behavior using the 1995–2011 Current Population Survey-Tobacco Use Supplement data. Our identification relies on the exploitation of the exogenous variations of the maximum UI weekly benefits across states and over years. Instead of focusing on all unemployed people, we concentrate on those who are eligible for UI benefits. We find that when the maximum UI weekly benefit level increases by $100, smoking cessation increases by approximately 2.9 percentage points among the UI-eligible unemployed. The results are robust to various model specifications. We also explore the mechanism and find suggestive evidence that the increase in the maximum UI weekly benefit leads to a decrease in the probability of overwork of the respective spouse. Moreover, the unemployed who are less educated are more responsive to the increasing UI benefits.

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1. Introduction

The smoking rate among the U.S. population has been declining since 1995, because the government has invested considerable efforts to control tobacco and related substance abuse. The current smoking rate among adults (aged 18 years or older) has declined from 20.9% in 2005 to 16.8% in 2014 (C.D.C, 2014)[1]. However, the trend varies significantly among labor force groups. Smoking prevalence among the unemployed is much higher than that among the employed (Fig. 1). Specifically, in 1995, approximately 36% of the unemployed and approximately 26% of the employed smoked. Although smoking prevalence in both groups has gradually decreased since 1995, smoking prevalence among the unemployed has reduced much more slowly than that among the employed. In 2011, the smoking prevalence rate among the unemployed was approximately 30%, nearly twice as high as that among the employed.

High smoking prevalence among the unemployed population has been paid increasing attention in recent decades. The literature provides substantial evidence for unemployment resulting in a high possibility for the unemployed to exhibit health-risk behaviors (Bolton and Rodriguez, 2009; Classen and Dunn, 2012; Deb et al., 2011; Falba et al., 2005; Kuhn et al., 2009; Montgomery et al., 1998), including hazardous drinking, excessive amount of cigarette consumption, and heavy dependence on illicit and prescription drugs[2]. The transition to unemployment implies a consequential future earning loss (Jacobson et al., 1993) and is associated with psychological distress (Tefft, 2011; Thomas et al., 2005). Cigarettes as well as alcohol and drugs are universally viewed as methods of dealing with stressful and uncontrollable social or economic problems for short-term pressure relief (Jarvis and Wardle, 1999)[3].

Hazardous health behaviors during unemployment impose remarkably negative impacts on the unemployed. As an apparent outcome, the unemployed are at a high risk of a deterioration in not only their physical health (e.g., a high risk of cardiovascular diseases (Walter et al., 2014), strokes (Gallo et al., 2004, 2006), and mortality (Browning and Heinesen, 2012; Clemens et al., 2009; Eliason and Storrie, 2009; Sullivan and Wachter, 2009)), but also mental dis-

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[1] http://www.cdc.gov/tobacco/data_statistics/fact_sheets/adult_data/cig_smoking/#states

[2] Henkel (2011) gives a detailed review of the literature concerning unemployment and substance use.

[3] Smoking can play as a pain reliever because of the embodied nicotine reduces tension and pressure (Jarvis, 2002).

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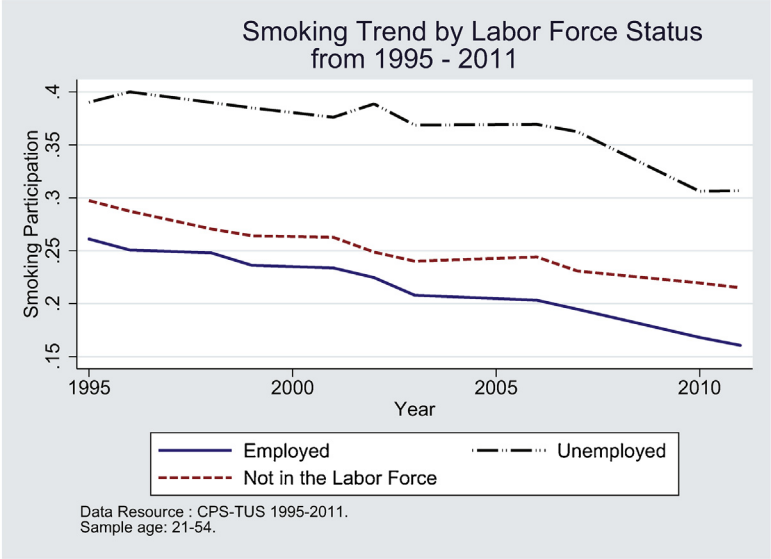

Fig. 1. Smoking Trend by Labor Force Status.from 1995 to 2011.

orders (Bartley, 1994; Gallo et.al, 2006; Mandal et al., 2011). This results in a substantial social cost. For the entire population, from 2000 to 2012, the U.S. tobacco-related health care cost amounted to $133 billion[4]. With an increase in the prevalence of smoking in the unemployed population, tobacco-related health care costs may also substantially increase in this population. Moreover, unemployment induces people, particularly young people, to establish life-long patterns of hazardous behaviors (Montgomery et al., 1998), forming a vicious cycle; these risk behaviors subsequently increase the likelihood of future unemployment and decrease the possibility of finding and holding down a job (Henkel, 2011).

Although smoking prevalence among the unemployed is a major concern, few government programs for smoking problems are specifically designed to target this population. In this paper, we use the unemployment insurance (UI) benefit system, the largest U.S. safety net program during the Great Recession (Bitler and Hoynes, 2016), as an example and substantiate whether increasing UI benefits reduces smoking among unemployed workers, which is a scarcely discussed topic in the literature on UI. We believe that this research offers valuable policy insights into the problem of smoking among unemployed workers.

This paper uses the data from the U.S. Current Population Survey-Tobacco Use Supplements (CPS-TUS) from 1995 to 2011, except 2000[5], to explore the effects of increasing UI benefits on the smoking behavior of the UI-eligible unemployed by exploiting the exogenous variations of maximum UI benefits across states and over the years. We merge the state maximum UI weekly benefit data with the CPS-TUS data. Consistent with the previous literature (e.g., Kenkel et al., 2014; Liu, 2010), we define two binary types of smoking behaviors: current smoking participation and smoking cessation over the past year. Considering the eligibility of the UI recipiency, we divide the unemployed sample into four groups: involuntary job losers, job leavers, those with recall expectations, and re-/new entrants (Krueger and Mueller, 2010). We investigate the effects of UI benefits on involuntary job losers who have been unemployed for less than 26 weeks[6] because they are UI-eligible and likely to receive UI benefits. They are supposed to be affected by the UI policy changes. We discuss the sample construction in the following sections.

Our empirical results reveal that the generous UI program can significantly increase smoking cessation among the UI-eligible unemployed. With the maximum UI weekly benefit level increasing by $100, smoking cessation can increase by approximately 2.9 percentage points. In addition, those who face more severe unemployment shocks benefit more from the state UI policy expansion. For example, less-educated unemployed individuals are more responsive to UI benefits, indicating that to a certain extent, the vulnerable population that needs social assistance the most may benefit more from the UI benefits. Moreover, the UI benefits expansion also exerts positive spillover effects on the spouse of the unemployed; specifically, when unemployed workers receive higher UI benefits, their spouses are less likely to over-work because the increasing UI benefits ease the family's financial pressure caused by the unemployment. Finally, we also discuss the potential sample selection bias. Our results are relatively robust according to a series of tests.

Our paper contributes to the literature in several aspects. First, in contrast to previous studies on smoking, which have used the entire unemployment population, we construct and focus on a more precise UI-eligible unemployed sample. The advantages of the UI-eligible sample allow us to directly estimate the health effects of UI on the UI-targeted population; thus, the effects would not be masked by the UI-ineligible population in the total unemployed. Second, this study is among the few to address the effects of UI on individuals' health behaviors. We consider the smoking behavior as an example, which provides an alternative insight into the association between health and the UI policy benefits. Third, we provide suggestive evidence for the UI alleviating financial pressure by comparing the heterogeneous effects of UI benefits across subgroups with different levels of financial pressure and by investigating the effects of UI on the over-work probability of the spouse.

[4] http://www.cancer.org/research/infographicgallery/tobacco-related-healthcare-costs

[5] They are CPS-TUS data in 1995sep, 1996jan 1996may, 1998sep, 1999jan, 1999may, 2001jun, 2001nov, 2002feb, 2003feb, 2003jun, 2003nov, 2006may, 2006aug, 2007jan, 2010may, 2010aug, 2011jan. We do not use TUS-CPS in 2000 because of the data limitations in that year.

[6] Those who are eligible for UI should fall into unemployment not through their own fault. And in the majority of states before the Great Recession, the maximum duration to receive UI is 26 weeks. We will discuss this in the data section.

Our paper is organized as follows: in Section 2, we provide a brief introduction of the state UI policy; in Section 3, we review the relevant literature; in Section 4, we describe the data sources and construction of the sample; in Section 5, we explain the identification strategy and model specification; in Section 6, we present the baseline results and robustness tests; in Section 7, we discuss the heterogeneous effects of UI benefits on the labor market outcomes of unemployed people's spouse; and in Section 8, we present the conclusion.

2. Unemployment insurance policy

The joint federal-state UI program was established as an essential part of the Social Security Act of 1935. The program is designed to provide eligible workers who are unemployed through no fault of their own with temporary cash benefits to ensure a basic standard of living during the severe unemployment period. The two major factors that determine people's UI eligibility and the amount of UI receipt are the reasons for their unemployment and regular work history (i.e., minimum earnings) over a predefined base period, which covers the earliest four of the previous five completed quarters before the job loss. Thus, the eligibility of receiving UI and the actual amount of UI benefits are related to individual's personal traits and past work status.

UI, as a joint federal-state program, allows state governments to design their own UI program following the general principles of the federal government. A state government can feasibly adjust the maximum UI weekly benefits and the maximum duration of UI receipt according to the state's own economic environment. This generates a considerable degree of heterogeneity in the UI policy across states and over years. Before the Great Recession, in the majority of states, the maximum duration of UI receipt was 26 weeks. MA and WA are the only two states that allow eligible people to receive UI benefits up to 30 weeks. The state maximum UI weekly benefits gradually increased before the Great Recession. Table 1 lists the state maximum UI weekly benefits in selected 4 years as examples. For instance, in MA, the nominal maximum UI weekly benefits increased from $496 in 1995 to $589 in 2006. During the prerecession period, all states had expanded their UI program in terms of the benefit level, although the duration remained almost constant.

During the Great Recession, starting from 2007, as the unemployment rate increased, the federal government funded the Extended Benefits (EB) program, which adds 13 weeks when the state's seasonally adjusted three-month moving average of the total unemployment rate (TUR) hit 6.5%. Some states even add more 7 weeks when the TUR exceeds 8%. Apart from the EB, since June 30, 2008, the federal government passed the Emergency Unemployment Compensation (EUC), which is a 100% federally funded program that provides benefits to individuals who have exhausted regular state benefits. Eventually, in many states, the maximum duration of UI receipt would range from 26 to 99 weeks depending on the severity of the crisis.[7] In addition to the UI duration expansion, the benefit level also upgraded remarkably during this period. As shown in Table 1, for example, state AK increased the maximum UI weekly benefits from $277 in 2006 to $370 in 2011.

In our study, we consider the expansion in the UI benefit level in each state during 1995–2011. The data used in our study are not suitable for us to exploit the variations in the duration of UI receipt. This is because smoking data from CPS-TUS cover a large sample in the pre-recession period during which variations in the UI program were only in terms of the benefit level, whereas the data cover only a two-year sample in the recession period.

[7] The EB expired on December 30, 2013 and the EUC on January 1, 2014.

Table 1
Unemployment Insurance Across States and Over Years (Selected 4 years).

state	1995	2001	2006	2011
AK	313	315	277	370
AL	266	241	245	265
AR	390	408	426	451
AZ	273	260	268	240
CA	339	292	502	450
CO	401	455	470	489
CT	494	504	519	555
DC	512	392	401	359
DE	443	400	368	330
FL	369	349	307	275
GA	303	348	346	330
HI	508	471	512	549
IA	331	347	362	376
ID	366	376	359	336
IL	357	376	391	388
IN	320	366	435	390
KS	384	406	416	435
KY	351	418	407	415
LA	267	328	288	247
MA	496	606	589	625
MD	369	356	379	430
ME	283	337	349	359
MI	432	381	404	362
MN	449	542	575	578
MO	258	298	301	320
MS	266	241	234	235
MT	337	334	386	421
NC	427	476	493	506
ND	359	372	379	442
NE	272	320	321	348
NH	301	382	415	427
NJ	522	545	581	598
NM	303	339	348	386
NV	350	370	386	398
NY	443	514	452	405
OH	362	367	383	387
OK	365	370	354	358
OR	444	478	484	496
PA	502	546	555	573
RI	478	504	532	551
SC	314	329	338	326
SD	266	285	297	314
TN	295	324	307	275
TX	372	373	375	415
UT	373	451	427	452
VA	307	340	368	378
VT	313	378	430	425
WA	517	560	553	570
WI	393	398	380	363
WV	428	415	424	424
WY	344	344	368	430

Data are collected from the BLS and are in 2011 dollars. In the whole sample, Min: 228; Max: 649; Mean: 393.87; SD: 86.35; Median: 381.

The heterogeneous variations in the maximum UI weekly benefits across states and over time provide economists with a valuable quasi-experiment (Gruber, 1997; Krueger and Mueller, 2010; Kuka, 2018; Walter et al., 2014). First, changes in the maximum UI weekly benefits, which are consequences of policy changes, affect the individual's actually received UI benefits but are not affected by individual's personal health status or the past work status (Walter et al., 2014). Thus, it can be viewed as exogenous if we control for the possibly related state characteristics; second, the maximum weekly benefit yields direct coefficient of policy interest than the individual's actually received UI benefits (Gruber, 1997). In this study, we use exogenous variations in the maximum UI weekly benefits in the state laws across states and over time to quantify the effect of UI generosity on smoking behaviors in the unemployed population.

3. Literature on unemployment benefits and health

In recent years, studies in the branch of health economics have focused on evaluating the health effects of increasing state UI benefits. An increase in UI benefits alleviates the financial shocks associated with job loss (Browning and Crossley, 2001; Kuka, 2018). In addition, it helps smoothen the consumption of the unemployed people (Gruber, 1997) and mitigate the loss of private health insurance due to unemployment (Brown, 2010). Consequently, the UI benefit program increases health insurance coverage and health utilization as well as improves the self-reported health (Cylus et al., 2015; Kuka, 2018) and mental health status (Tefft, 2011), thereby reducing suicide propensity (Cylus et al., 2014).

We extend on the landmark studies that have provided new insights into the social benefits of the UI policy in the following aspects:

If we identify the UI-eligible unemployed workers and focus on them instead of the entire unemployed workers population, we are more likely to estimate the actual effects of UI benefits. The health literature has typically considered the entire unemployed population as their target sample and has encountered occasional failures in verifying the possible underlying substantial influence of UI because the real effects on the UI-eligible group are likely masked by the large UI-ineligible group. For example, Kuka (2018) quantifies UI benefits on several health outcomes, including health insurance coverage, utilization, and some health-risk behaviors, by using the Behavioral Risk Factor Surveillance System dataset. Although the results of Kuka (2018) reveal that UI benefits increase health insurance coverage and health utilization among unemployed workers, no evidence suggests that UI benefits affect health-risk behaviors, such as smoking and drinking. Kuka (2018) admits to a certain bias in the results because the types of unemployment are not identified in the study sample. By contrast, Krueger and Mueller, (2010)do not observe significant influences of UI benefits on job search intensity among the entire unemployed sample, and they report significant negative effects of UI benefits on the UI-eligible subsample. Studies such as those of Kuka (2018) and Krueger and Mueller, (2010) have highlighted the need for a UI-eligible sample. In this paper, we classify unemployment into four types (involuntary job losers, job leavers, those with recall expectations, and re-/new entrants) and focus on the involuntary job losers, who are most likely to be eligible for UI benefits. Considering the UI benefit receipt duration, our final UI-eligible sample comprises job losers with an unemployment duration of less than or equal to 26 weeks. We will discuss our method to construct the UI-eligible sample in the data section.

Apart from the aforementioned studies that have focused on the causal relationship of UI benefits and health outcomes, few studies have explored the health effects of UI on the determinants of individual's health status, such as excessive smoking or drinking behaviors. We believe that analyzing the effects of UI on people's health behaviors is essential. Policymakers should understand not only the outcomes but also the implementation of the final policy. Health status is the outcome of health behaviors. Understanding how individual's health behaviors are affected by UI benefits can reveal certain mechanisms that explain the association between a social assistance program and health status. Furthermore, smoking and excessive drinking behaviors not only reflect the potential risks in physical health but also act as a symptom, reflecting the psychosocial stress suffered by the unemployed (Vogli and Santinello, 2005); discerning the feelings of the unemployed after losing their jobs is difficult, but their smoking or drinking behaviors can reflect the unobservable pressure, which enables us to quantify the psychological effects of UI benefits. Thus, advancing from the general health status, we explore the effects of increasing UI benefits on one typical health behavior—smoking—when individuals face unemployment.

The rationale that UI benefits reduce smoking is simple and mainly stems from the "pressure-smoothing" feature of smoking. Psychologists, such as Bradstock et al., (1988) and Metcalfe et al. (2003), have identified that higher levels of pressure are closely related to a larger amount of cigarette consumption. In addition, considering the demand of smoking as a "pleasurable, pressure-smoothing" activity is rational because the transition to unemployment implies a substantial future earning loss (Jacobson et al., 1993) and is associated with psychological distress (Tefft, 2011; Thomas et al., 2005); moreover, smoking cigarettes, drinking alcohol, and using drugs are universally viewed as practices of dealing with stressful and uncontrollable socioeconomic problems for short-term pressure relief (Jarvis and Wardle, 1999)[8]. For example, the empirical results of Carpenter et al. (2017) reveal that the use of prescription pain relievers and hallucinogens increases when people face substantial shocks during economic downturns. Nevertheless, increasing UI benefits can alleviate the financial shock associated with job loss (Browning and Crossley, 2001; Kuka, 2018), which might reduce the possibility of smoking through the psychological channel.

Assuming the pressure-smoothing effects of UI benefits, we conclude that those who suffer severe psychological distress benefit more from the increase in UI benefits. Because few studies have investigated this hypothesis, we test whether the UI benefits are well-defined to target the population that needs the most help.

4. Data

In this section, we present the specifications of the UI-eligible sample and the definition of smoking measurements and other major variables used in our analysis.

4.1. Data sources

Our research is specifically designed for the UI-eligible sample as well as the measurements of smoking behavior. The primary data set adopted is 1995–2011 CPS-TUS.

The CPS-TUS is a universally used, nationally representative cross-sectional dataset used for tobacco control research. It is conducted by the US Census Bureau; it was initially sponsored by the National Cancer Institute, but since 2001, it has been cosponsored by the Centers for Disease Control and Prevention. In each wave, it covers more than 100,000 respondents aged 15 years and older. It investigates information on their smoking characteristics, such as current and one-year-ago smoking status, which enables us to measure the static and dynamic smoking behaviors. It can also be linked with monthly CPS data, and detailed demographic background and employment information can be extracted to define our UI-eligible sample.

Other data sets, such as that of the Panel Study of Income Dynamics (PSID), are not compatible with our research design. Although the PSID provides detailed information on smoking behavior, it does not investigate the reasons for unemployment; therefore, we cannot identify the UI-eligible group. Some administrative data, such as the labor unemployment registration data, document the UI eligibility and amount of actual UI benefits but lack information on smoking behavior.

We obtain the year-specific maximum UI weekly benefit data from the U.S. Department of Labor's Comparison of State UI Laws[9]. We then merge the state maximum UI weekly benefits with the CPS-TUS on states and years.

[8] Smoking can play as a pain reliever because of the embodied nicotine reduces tension and pressure (Jarvis, 2002).

[9] http://workforcesecurity.doleta.gov/unemploy/statelaws.asp#Statelaw.

We restrict our sample to unemployed people aged from 21 to 54 years[10]. First, regular work history is required for people to be eligible for receiving UI. For example, UI claims data in August 2011 show that people younger than 22 years account for only 2.8% of total claims[11]. The population aged 22–54 years occupies the majority of claims, accounting for approximately 76.1% of total claims. Second, since 2014, an increasing number of states have implemented the Tobacco 21 policy, which forbids teenagers under the age of 21 years from buying various tobacco products. To strengthen the implication of the policy in our paper, we set the lower bound of age at 21. Third, since 2007, the labor force participation rate in the population aged 55 years is maintained at a low level at approximately 40%, and the population aged between 21 and 54 years contributes most to the US labor workforce[12].

4.2. Specifications of UI-eligible sample

In practice, it is the most accurate method to separate the UI-eligible sample from the total unemployed population by separately checking all the qualification terms. However, this ideal approach is not feasible with survey data because the qualification terms are complex and the information provided by the survey data is incomplete. Therefore, we mainly specify the UI-eligible sample using two major criteria: whether unemployed individuals lose their jobs through their own faults and the duration of a current unemployment spell.

The TUS-CPS survey questions each unemployed individual on their reasons for unemployment and classifies them as one of the following six groups: job loser (on layoff), other job loser, temporary job ended, job leaver, re-entrant, and new entrant; moreover, the survey enquires whether they are given a date by the employer to return to work. Similar to Krueger and Mueller, (2010), we reclassify the unemployed into the following four subgroups:

Temporary layoffs with recall expectations

They belong to the following unemployment types: job losers (on layoff), other job losers, and temporary job ended. They also report to **have been** given an indication **to** be recalled to work by the former employers in the following 6 months.

Job losers

They belong to the following unemployment types: job losers (on layoff), other job losers, and the temporary job ended. They report to **have not been** given the indication to be recalled to work by the former employers in the following 6 months.

Job leavers

They report being voluntary job leavers in the CPS-TUS survey.

New/re-entrants into the labor market

They report to be re-entrants or new entrants into the new labor markets.

Our baseline sample comprises only job losers because they apparently satisfy the "not through own fault" criterion for UI claims. We exclude the re-entrants and new entrants because they do not meet the earning and work time criteria in a given "base period." We also exclude the job leavers because they are unemployed through their own volition. Whether to include the temporary unemployed with recall expectations is ambiguous because it is not obvious whether they quit their jobs voluntarily. We follow Krueger and Mueller, (2010) and do not include this group into our sample. In the empirical section, we also construct a larger sample by pooling the job losers with those with recall expectations to test the robustness of the baseline results

Data from the CPS-Unemployment Insurance Supplements Survey in 2005 (Vroman, 2009) provides us evidence that job losers are the most likely to claim and receive UI benefits. In Appendix Table A1[13], approximately 50.7% of the total job losers applied for the UI benefits, which is the highest percentage and almost three times higher than the second group. The UI-benefit taking-up rate among the job losers amounts to 35.6%, whereas it is only 8.8% among the job leavers. The application rates and taking-up rates confirm that job losers are the dominant group in the UI-eligible population.

We also limit our sample to those who are in an unemployment spell of less than 26 weeks. Most of the states set the maximum duration for receiving UI benefits to 26 weeks before the Great Recession. We cautiously use the 26-weeks sample because people's behaviors, such as that of job seeking, are observed to change significantly after the exhaustion of UI (Krueger and Mueller, 2010). As a robustness test, we also relax our unemployment requirement to 30 weeks because people require time to claim the UI benefits and the maximum duration is 30 weeks in some states. The 26-week sample allows us to substantiate the short-term effects of UI but not the long-term effects.

Finally, whether the UI benefits system covers the part-time job varies from state to state. We also consider this policy difference across states in our sample construction. In summary, we focus on the UI-eligible sample, comprising job losers in an unemployment spell of less than 26 weeks.

To sum up, the unemployed group, according to the TUS-CPS, can be classified into four categories. They are temporary layoffs with recall expectations, job losers, job leavers, and new/re-entrants into the labor market. In this paper, the group of job losers is referred to as the UI-eligible-type unemployed; the other three categories of the unemployed are, as a whole, referred to as the UI-ineligible-type unemployed. In our regression analysis, we focus on the UI-eligible sample, which consists of job losers who are unemployed for less than 26 weeks. Simply speaking, our UI-eligible sample requires two conditions: the unemployed must be the UI-eligible type and their unemployment spell must be less than 26 weeks.

4.3. Major dependent variable: smoking status

We examine two major outcomes reflecting individual's smoking status— smoking participation and smoking cessation—identically defined by Kenkel et al. (2014) and Liu (2010). Participation is simply a static description of the smoking status at the survey time, and cessation is similar to a dynamic depiction if we consider the decision to smoke today as a reflection of the accumulated decisions on initiation[14] and cessation in the past years (DeCicca et al., 2008; Liu, 2010).

The CPS-TUS respondents are asked a series of questions on their current smoking status, smoking history, and attitude toward smoking. For example, CPS-TUS questions include, "(Q1) Have you ever smoked 100 cigarettes in your lifetime?"; those who respond with a "Yes" to Q1 are asked the following two questions: "(Q2) Do you now smoke cigarettes every day, some days, or not at all?" and "(Q3) Did you smoke cigarettes every day, some days, or not at all around 12 months ago?" Based on responses to these 3 questions, we develop smoking participation and cessation indicators.

[10] Our results become weaker if we include individuals aged over 55 years into our sample. This may be because those aged 55 years old could withdraw their pension from their 401K account without penalty, which can reduce the income shock due to job loss.

[11] https://workforcesecurity.doleta.gov/unemploy/content//chariu2011/2011Aug.html

[12] https://www.bls.gov/opub/ted/2017/employment-population-ratio-and-labor-force-participation-rate-by-age.htm

[13] We integrate the statistics from Vroman (2009).

[14] In the U.S., most smoking initiation happens before 18 years old. Our sample is the unemployed workers who are more than 20 years old. Therefore, initiation behavior is not our study outcome.

Respondents whose response is "No" for Q1 or "not at all" for Q2 are classified as nonsmokers, coded 0 for smoking participation. Respondents whose response is "now smoke cigarettes every day or some days" to Q2 are classified as smokers, coded as 1 for smoking participation.

To define smoking cessation, we combine the responses to Q2 and Q3. Conditional on being every day or occasional smokers around 12 months ago, if the respondents continue to remain in the everyday or occasional smoker categories, they are coded as 0 for smoking cessation. If they respond with "smoke not at all" at present, they are coded as 1 for smoking cessation.

4.4. State-level data resources

The data of state social programs are directly obtained or calculated from the UKCPR National Welfare Data, collected by The University of Kentucky Center for Poverty Research (UKCPR, 2016). The state smoke-free air law data are obtained from the ImpacTeen dataset before 2008 and from the CDC State dataset after 2008. Following DeCicca et al. (2008), we calculate the state anti-smoking sentiment (SASS) for each state. In the 2003 and 2010/2011 waves of CPS-TUS, we impute the SASS because the questions we need to construct the SASS are not surveyed. We assume the SASS in these years to be the same as that in the previous year when it is available. In our regression analysis, we also add a dummy variable indicating if the observation has an imputed value for this variable.

4.5. Observations with missing data

Some of the respondents do not report the number of children. We do not want to exclude these observations because our sample size is limited. Observations with missing children number for respondents who are unmarried and married are set to have the average of the children number in the unmarried and married household populations, respectively. These averages are rounded to integers. In our regression, we add a dummy variable to indicate observations with missing number of children.

The inclusion of the variable "the fraction of the state house that is democrat" is missing for Nebraska. To handle this missing value, we impute the value as 0.5 because Nebraska has a uniquely, officially unicameral and nonpartisan legislature, different from all other states. We also include a dummy to indicate a state as Nebraska in the regression to take care of this imputation. We also have other state characteristics missing for the District of Columbia. Considering it is not a state in the United States and is different from other states in terms of the political environment and economic policy, we drop the observations from the District of Columbia to handle the missing values of state characteristics.

5. Models and identification strategy

Our empirical identification strategy relies on the exploration of the exogenous variations of maximum UI weekly benefits across states and within state over the years, similar to Brown (2010); Krueger and Mueller, (2010); Gruber (1997), and Kuka (2018)[15].

Considering that individual smoking behavior is a binary variable, we construct a probit model to estimate the effect of the state maximum UI weekly benefits on the probability of being a current smoker (participation model) and the probability of quitting smoking over the past year (cessation model). The baseline model is as follows:

$$Smoking_{ist} = \beta_0 + \beta_1 UI_{st} + \beta_2 X_{ist} + \beta_3 Z_{st} + \delta_t + \psi_s + t\psi_s + \varepsilon_{st} \quad (1)$$

$Smoking_{ist}$ represents the smoking outcomes, namely participation and cessation, of individual i at state s and time t. In the participation model, $Smoking_{ist} = 1$, implying that the individual is a current smoker, which is otherwise 0. In the cessation model, $Smoking_{ist} = 1$, indicating that the individual quit smoking over the past year, which is otherwise 0.

We focus on a major dependent variable, UI_{st}, which is the maximum UI weekly benefits in state s in time t. X_{ist} is a vector for individuals' demographic characteristics, such as age groups, gender, race, education level, and the number of children (see variables in Table 2). We also include a dummy variable to indicate whether the respondents have missed reporting their number of children.

Z_{st} is a vector representing state's economic environment. First, it includes the state poverty rate, the state unemployment rate, the state-year wage dispersion index[16], the log of the state GDP per capita, the state proportion of population with Bachelor's degree or more, whether the state governor is democratic, and the fraction of the state house that is democrat. We also include the log of the state real cigarette price and state anti-smoking sentiment (SASS, DeCicca et al., 2008) to control for the state anti-smoking environment. In our robustness tests, we add the maximum UI weeks and other state safety net variables, which include the following: the state minimum wage; state Supplemental Nutrition Assistance Program (SNAP, formerly known as food stamps) benefit for two-, three-, and four-person families; state-earned income tax credit (EITC) refundable status; EITC maximum credit for two and three dependents; state WIC program participation rate; Medicaid enrollment rate; AFTC participation rate; maximum social security income per person[17]; and state union coverage[18]. We also control for other state smoking-related policies, which include the log of state tobacco control spending and binary variables for four categories of state smoke-free air laws[19], namely "no bans," "weak bans," "moderate bans," and "severe bans."

Finally, δ_t are sets of year and month dummies that control for the time fixed effects (FEs). ψ_s are sets of state dummies that control for the state time-invariant effects. $t\psi_s$ captures the state-specific linear time trend.

Standard errors are clustered at the state level to capture the possible correlations between individuals within the state.

6. Empirical results

6.1. Correlation between UI and individuals' smoking behaviors

Fig. 2 presents the trend during our study period for the unemployed workers' smoking participation, and smoking cessation as well as the real maximum UI weekly benefits. With the increasing UI benefits in this period, the prevalence and intensity of smoking experienced a noticeable reduction. In particular, the UI

[15] Studies like Krueger and Mueller, 2010utilize the variations in the UI benefits level, and studies like Gruber (1997) and Kuka (2018) construct the simulated income replacement ratio to measure the UI benefits.

[16] We follow Krueger and Mueller, 2010to create the state-year wage dispersion index. The state wage dispersion index is calculated using a wage equation that includes age, age square, gender, education level and year, and state dummies (Krueger and Mueller, 2010). In the wage regression, we use a sample of 3,032,430 workers from the CPS outgoing rotation group files from 1995 to 2014. Then we use the standard deviation of residuals for each state-year from the wage equation as the indicator of state wage dispersion. We only use the state wage dispersion indicators up to 2011 in our UI analysis.

[17] These variable are from UKCPR National Welfare Data, 1980-2015, http://www.ukcpr.org/data

[18] http://www.unionstats.com/

[19] The data are from ImpacTeen (http://impacteen.uic.edu/tobaccodata.htm) for the years before 2008, and from CDC STATE dataset for the years after 2008.

Table 2
Summary Statistics by Labor Force Status.

	(1) Employed	(2) Not in Labor Force	(3) Unemployed	(4)	(5)	(6)	(7)
			Total Unemployed	Recall Expectations	Job Losers	Job Leavers	New/ Re-Entrants
smoking participation	0.218	0.254	0.362	0.359	0.368	0.407	0.333
	(0.413)	(0.435)	(0.481)	(0.480)	(0.482)	(0.491)	(0.471)
smoking cessation	0.0858	0.0762	0.0705	0.0839	0.0637	0.0678	0.0809
	(0.280)	(0.265)	(0.256)	(0.277)	(0.244)	(0.251)	(0.273)
age:21-25	0.115	0.157	0.205	0.143	0.146	0.229	0.319
	(0.319)	(0.364)	(0.404)	(0.350)	(0.353)	(0.420)	(0.466)
age:26-30	0.135	0.130	0.160	0.150	0.145	0.179	0.180
	(0.342)	(0.336)	(0.366)	(0.357)	(0.352)	(0.383)	(0.385)
age:31-35	0.151	0.142	0.148	0.158	0.151	0.150	0.140
	(0.358)	(0.349)	(0.355)	(0.365)	(0.358)	(0.357)	(0.347)
age:36-40	0.164	0.146	0.144	0.161	0.155	0.142	0.120
	(0.370)	(0.353)	(0.351)	(0.368)	(0.362)	(0.349)	(0.325)
age:41-45	0.167	0.144	0.137	0.152	0.155	0.123	0.106
	(0.373)	(0.351)	(0.344)	(0.359)	(0.362)	(0.329)	(0.308)
age:46-50	0.159	0.149	0.124	0.136	0.149	0.106	0.0834
	(0.366)	(0.356)	(0.330)	(0.343)	(0.356)	(0.307)	(0.277)
age:51-54	0.109	0.131	0.0823	0.0986	0.0990	0.0714	0.0518
	(0.312)	(0.338)	(0.275)	(0.298)	(0.299)	(0.258)	(0.222)
less than high school	0.0825	0.212	0.187	0.208	0.179	0.154	0.208
	(0.275)	(0.409)	(0.390)	(0.406)	(0.383)	(0.361)	(0.406)
high school	0.302	0.330	0.370	0.415	0.390	0.380	0.319
	(0.459)	(0.470)	(0.483)	(0.493)	(0.488)	(0.485)	(0.466)
more than high school	0.300	0.275	0.279	0.257	0.269	0.276	0.304
	(0.458)	(0.447)	(0.448)	(0.437)	(0.443)	(0.447)	(0.460)
university and above	0.316	0.183	0.164	0.120	0.162	0.190	0.169
	(0.465)	(0.386)	(0.370)	(0.325)	(0.369)	(0.392)	(0.375)
female	0.479	0.736	0.471	0.393	0.388	0.519	0.623
	(0.500)	(0.441)	(0.499)	(0.488)	(0.487)	(0.500)	(0.485)
black	0.0839	0.116	0.170	0.0932	0.166	0.133	0.212
	(0.277)	(0.321)	(0.375)	(0.291)	(0.372)	(0.340)	(0.409)
other races	0.0580	0.0820	0.0753	0.0564	0.0665	0.0714	0.0978
	(0.234)	(0.274)	(0.264)	(0.231)	(0.249)	(0.258)	(0.297)
unmarried	0.392	0.427	0.596	0.471	0.578	0.596	0.664
	(0.488)	(0.495)	(0.491)	(0.499)	(0.494)	(0.491)	(0.472)
# of children: 0	0.204	0.190	0.234	0.216	0.268	0.194	0.195
	(0.403)	(0.392)	(0.424)	(0.411)	(0.443)	(0.395)	(0.396)
# of children: 1	0.617	0.600	0.598	0.591	0.563	0.666	0.635
	(0.486)	(0.490)	(0.490)	(0.492)	(0.496)	(0.472)	(0.481)
# of children: 2	0.120	0.120	0.104	0.119	0.106	0.0884	0.102
	(0.325)	(0.325)	(0.305)	(0.323)	(0.308)	(0.284)	(0.302)
# of children: 3	0.0440	0.0603	0.0430	0.0530	0.0431	0.0356	0.0428
	(0.205)	(0.238)	(0.203)	(0.224)	(0.203)	(0.185)	(0.202)
# of children: more than 4	0.0149	0.0295	0.0210	0.0213	0.0194	0.0167	0.0254
	(0.121)	(0.169)	(0.143)	(0.144)	(0.138)	(0.128)	(0.157)
# of children is missing	0.494	0.489	0.478	0.473	0.436	0.552	0.527
	(0.500)	(0.500)	(0.500)	(0.499)	(0.496)	(0.497)	(0.499)
Observations	688706	138129	37204	2960	19352	4186	10706

The observation number for cessation is 119,779 in employed group, 11,026 in the unemployed group, and 29,295 in the Not-in- Labor-Force group. Standard deviations are in parentheses.

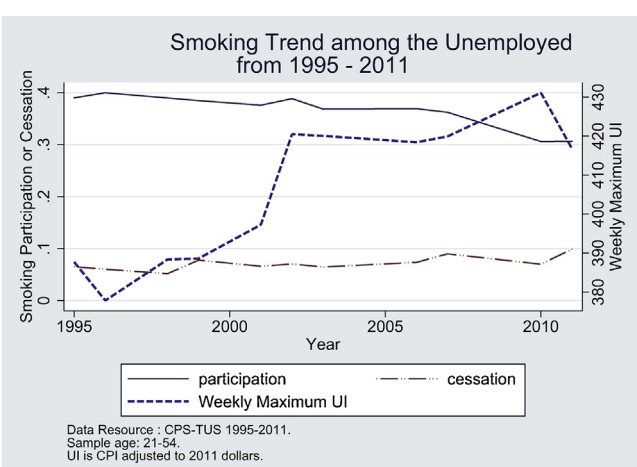

Fig. 2. Smoking Trend among the Unemployed.from 1995 to 2011.

average benefits (adjusted to dollars in 2011) increased from $385 in 1995 to $418 in 2011. In addition, smoking prevalence among the unemployed declined sharply from 39.03% in 1995 to 30.66% in 2011. Furthermore, the smoking cessation possibility increased from 6.52% in 1995 to 9.88% in 2011.

The smoking status also varies by different unemployment groups, namely, those with recall expectations, job losers, job leavers, and re-/new entrants. The Fig. 3 presents the trend for smoking participation by unemployment groups. It highlights three aspects: (1) The overall smoking prevalence has a downward slope even for the UI-ineligible typed unemployed, such as job leavers, reminding us to be cautious about the interpretations of the systematical synchronization of the UI benefits and smoking trends shown in Fig. 2. (2) The unemployed who experience greater pressure are more likely to smoke than those who experience lower pressure. For example, job losers and job leavers face much severe unemployment shock compared with those with recall expectations; smoking prevalence is higher among job losers

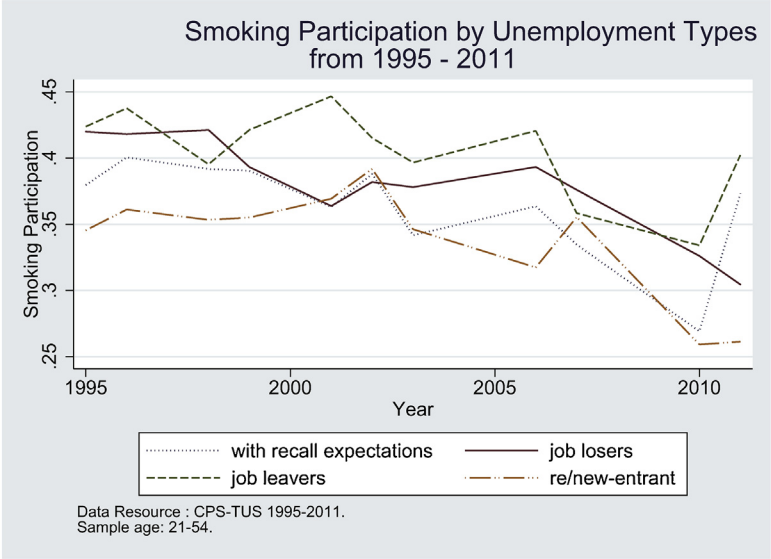

Fig. 3. Smoking Participation by Unemployment Types.from 1995 to 2011.

and leavers than among those with recall expectations. (3) Job losers (UI-eligible) smoke less than the job leavers (UI-ineligible type), suggesting that UI benefits may be positively related to smoking reduction. Compared with the job losers, the job leavers should experience less unemployment shocks because they voluntarily quit their jobs; however, they still form a higher proportion of the smoking population. Workers who quit their job are possibly impulsive, with higher discount rates, and more likely to smoke. The lower smoking prevalence among job losers might be attributed to the increasing levels of UI benefits, which is also our main concern in this paper.

6.2. Basic data description

Table 2 reports the descriptive statistics of smoking outcomes and demographic characteristics for the employed, unemployed, and those not in the labor force. Compared with the employed, the unemployed are more likely to be associated with characteristics such as young age, minority, unmarried, and less education. In terms of smoking outcomes in all three groups, the unemployed have the highest smoking prevalence (36.2%) and the lowest smoking cessation rate (approximately 7.1%). This result is consistent with those of previous studies: people under unemployment shocks have a high possibility of indulging in health-risk behaviors (Bolton and Rodriguez, 2009; Deb et al., 2011; Falba et al., 2005).

We further divide the unemployed into four groups, namely those with recall expectations, job losers, job leavers, and new/re-entrants. Those with recall expectations appear to be similar in general to job losers; however, approximately 15% of the unemployed with recall expectations are from the minority race, which is approximately 24% of job losers. Among these two groups, job leavers tend to be younger and more educated. In addition, the female population occupies more than 50% of the job leavers group. Focusing on smoking behaviors, we find that smoking prevalence among job losers (UI-eligible type) is lower than that among job leavers (UI-ineligible type), although no evident difference is observed for smoking cessation between them.

6.3. Effects of UI on smoking

We first compare the estimates of the entire unemployed sample with those of the constructed UI-eligible unemployed sample. Table 3 reports the marginal effects of the maximum UI weekly benefits on smoking behaviors for participation in columns 1–4 and for cessation in columns 5–8.

In columns 1 and 5, we analyze the effect of UI among the entire unemployed sample, controlling for unemployment types in addition to other demographic controls (see the notes in Table 3). We also control for the year and month FEs. Estimates suggest no significant effects of the UI benefits on smoking cessation of the total unemployed population. A $100 increase in UI benefits leads to a decrease of 1.9 percentage points in the smoking participation among the total unemployed population. This sample comprises both the UI-eligible and UI-ineligible typed unemployed, which possibly leads to some measurement errors (Kuka, 2018). The presence of measurement errors attenuates our estimates toward zero. Assuming that the UI-ineligible type is not affected or less affected by the UI benefits, the values in columns 1 and 4 are underestimated. This implies that an accurate analysis of the UI-eligible sample is necessary.

Starting from columns 2 and 6, we narrow our sample to job losers who are the UI-eligible-unemployed type with an unemployment spell of less than 26 weeks (hereafter, UI-eligible sample). We restrict the unemployment duration to be no more than 26 weeks because in the majority of states, people are eligible for receiving UI benefits for a maximum of 26 weeks. The estimates reveal that UI benefits are helpful to prevent smoking among the UI-eligible unemployed workers. Specifically, a $100 increase in the weekly UI benefit level reduces the smoking participation by 2.5 percentage points and enhances the smoking cessation by 1.3 percentage points. Considering that the sample means of smoking participation and cessation for job losers are 36.8% and 6.37%, respectively, a $100 increase in the state maximum UI benefit level can lower smoking participation by 6.8% and increase smoking cessation by 20%.

To be cautious, the results so far are threatened by the omitted variable problem. If omitted variables, especially macroeconomic conditions, affect people's smoking behaviors and are correlated with state UI benefits, our estimates are inconsistently estimated. For example, the deteriorating economic environment, such as high state unemployment rates, may lead unemployed people to smoke more. Furthermore, if the unemployment rates continue to increase, state governments have the incentives to increase the UI benefits; this is representative of the events during the 2008 financial crisis. Similarly, if state governments increase cigarette taxes to obtain the revenue to compensate for the increased UI benefits, the estimated effects of UI benefits may reflect the effects of cigarette

Table 3
Effect of UI on Smoking Participation and Cessation – Baseline.

	(1) Participation	(2)	(3)	(4)	(5) Cessation	(6)	(7)	(8)
VARIABLES	Total Unemployed	UI Eligible Sample			Total Unemployed	UI Eligible Sample		
UI ($100)	−0.019**	−0.025***	−0.017***	−0.027**	0.005	0.013***	0.015***	0.024**
	(0.007)	(0.008)	(0.006)	(0.010)	(0.003)	(0.004)	(0.005)	(0.012)
Observations	37,204	13,800	13,800	13,800	11,026	4,258	4,258	4,177
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics			Yes	Yes			Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE				Yes				Yes

Table 4
Effect of UI on Smoking -Control Other Policy Environment.

VARIABLES	(1) Participation	(2)	(3)	(4)	(5)	(6) Cessation	(7)	(8)	(9)	(10)
UI ($100)	−0.027**	−0.027***	−0.022**	−0.021*	−0.016	0.024**	0.022*	0.028**	0.027**	0.029**
	(0.010)	(0.010)	(0.010)	(0.011)	(0.016)	(0.012)	(0.012)	(0.013)	(0.014)	(0.014)
UI maximum weeks		0.000	0.000	0.000	−0.000		−0.001	−0.001	−0.001	−0.001
length		(0.001)	(0.001)	(0.001)	(0.002)		(0.001)	(0.001)	(0.001)	(0.002)
Observations	13,800	13,800	13,800	13,800	13,800	4,177	4,177	4,177	4,177	4,177
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length		Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes
Other Welfare Policies			Yes	Yes	Yes			Yes	Yes	Yes
Smoking Policies				Yes	Yes				Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend					Yes					Yes

taxes on smoking behaviors. Nevertheless, these omitted variables are likely to bias our results. To probe the robustness of results, we impose more stringent controls in columns 3, 4, 7, and 8.

In columns 3 and 7, we further control for the state economic environment, including the state poverty rate, log of the state GDP per capita, state-year wage dispersion index, the state proportion of population with Bachelor's degree or more, and unemployment rate. To capture the state policy-making environment, we control for the democratic status of the state governor and the fraction of state house that constitutes democrats. We also add the state anti-smoking environment, including the log of the real cigarette price and SAAS. If any observation has an imputed value for one variable, we add a dummy variable indicating the imputation for that variable. The impacts of UI benefits on smoking participation and cessation do not considerably change compared with those in columns 2 and 5.

In columns 5 and 8, considering the unobserved state characteristics, we additionally control for the state FEs. A $100 increase in the state maximum UI weekly benefit level reduces smoking participation by 2.7 percentage points and increases smoking cessation by 2.4 percentage points. These findings are stronger than the impacts estimated in the undemanding regression model in columns 2 and 5. This reveals that the omitted state macro-environment in undemanding regression tend to be positively correlated with the state UI benefit level and negatively correlated with smoking behaviors of unemployed workers.

6.4. Control for confounding state policy environments

In this section, we test whether our results are sensitive to the inclusion of other state safety net policies in our model.

First, UI expansions occur in terms of not only the benefit level but also the time duration. To distinguish the impacts of benefit expansion from those of duration expansion, in Table 4, we add the duration of UI in addition to the controls in Table 3. For the convenience of comparison, the estimates reported in columns 1 and 6 are copied from columns 4 and 8 of Table 3. Estimates in columns 2 and 7 are almost unchanged after including the control for the state UI maximum week length. This could be attributed to the fact that the CPS-TUS survey was implemented in only 2010 and 2011 during the recession period and covered a small proportion of the sample in our study. Our estimates are mainly driven by the impacts of the UI benefit level expansion.

Another concern is that during the Great Recession, the government tended to expand safety nets to provide emergent social assistants. As shown in Table 4, although the state FEs control for time-variant factors, we are still concerned regarding other time-variant welfare policies occurring concurrently with UI expansion, which may affect the health behaviors of unemployed people. Therefore, in columns 3 and 8, we additionally control for other representative social welfare policies; for example, the minimum wage, the food stamp policy, the state EITC, SSI, WIC, AFTC, and Medicaid. We also control for the state union coverage to handle the welfare from the workers' unions. The marginal effects of the UI benefit level are −0.022 on smoking participation and 0.028 on smoking cessation, which are consistent in both magnitude and significance level.

We also suspect that ignoring other state anti-smoking policies might lead our estimates to be overestimated if some state anti-smoking policies possibly correlate with the UI policy through certain unobservable factors. In columns 4 and 9, we continue to control for other state-level anti-smoking policies, namely the log of state tobacco control spending and a set of binary variables for categories of state smoke-free air laws (no bans, weak bans, moderate bans, and severe bans). These estimates still preserve the magnitude and significance.

Even though we control for a considerable set of state economic and policy environment variables as well as state FEs and year

FEs, these efforts could be insufficient as differential time trends in states could still be a concern that drives our main results. For example, as shown in Fig. 2, a substantial strike exists in the UI benefit between 2000 and 2003. In addition, the PRWORA and TANF programs passed in the late 1990s resulted in major changes in unemployment benefits. To address this concern, we further attempt to control for the state-specific linear time trend to capture the differential linear time trend in each state. In columns 5 and 10, after adding the trend, we lose the precision of the estimate for smoking participation. The magnitude is also smaller than the effect in other columns. The loss of precision could be attributed to the large variation absorbed by the state FEs, year FEs, and state-specific linear trend. If we regress state UI benefits on these FEs and trend, R^2 would be more than 0.90. We still achieve a relatively consistent estimate for smoking cessation in column 10.

6.5. Robustness tests

6.5.1. Robustness to changes in UI measurements and sample specifications

Some unemployed workers could possibly care about the percentage change in UI benefits instead of UI benefit levels. Therefore, we use log UI as the key independent variable in Panel A in Appendix Table A2.

Kuka (2018) uses a simulated income replacement rate to measure UI benefits. The replacement rate is defined as the UI payment as a share of pre-unemployment weekly earnings. Following Kuka's approach[20], we obtain two UI replacement ratios from the United States Department of Labor[21]. The department collects the actual amount of weekly UI benefits (WBA) that individuals receive and their hourly wage based on their claims. Replacement Ratio-1 is the weighted average of (WBA/normal hourly wage × 40 h). Replacement Ratio-2 is (weighted average WBA)/(weighted average normal hourly wage × 40 h). If these two replacement ratios perform in the same way as the generosity measure that Kuka (2018) generated, then our replication would provide some useful insights. The results are reported in Panels B and C in Appendix Table A2. These two replacement ratios can only be traced back to 1997; thus, the observations in 1995 and 1996 are missing. In this replication, we can still obtain significant results for smoking cessation, although not for smoking participation.

Until now we have limited our sample to job losers with employment duration less than 26 weeks. To test the robustness of our results, we relax these restrictions to examine whether our baseline results remain consistent.

First, we extend our sample to those unemployed for 30 weeks because claiming the UI benefits requires time[22] and the maximum unemployment duration is 30 weeks in some states. In Panel A of Appendix Table A3, the marginal effects of the maximum UI weekly benefits correspond to our baseline results in both magnitude and significance. In Panel B, even after we relax the restriction on unemployment duration to less than one year (52 weeks), the results are consistent.

We also reconfirm the strength of our results by including those unemployed individuals with recall expectations from their former employers into our sample because compared with job leavers and re-/new entrants, it is ambiguous whether those with recall expectations lost their jobs through their own fault. In Panel C of Appendix Table A3, the impacts of UI benefits are slightly weaker than those shown in Table 4. One possible reason could be that the unemployed with recall expectations are not as eligible as the job losers; thus, fewer of them benefit from UI benefits.

Table 5
Effect of UI on Smoking Behaviors 12 Months Ago.

VARIABLES	(1) 12-month-ago Participation	(2)	(3) 12-month-ago Cessation	(4)
UI ($100)	−0.017	−0.003	0.011	−0.010
	(0.013)	(0.021)	(0.012)	(0.017)
Observations	12,455	12,455	4,048	4,048
Demographics	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes

6.5.2. Placebo test: smoking behaviors 12 months ago

The literature (Krueger and Mueller, 2010; Walter et al., 2014) considers the changes in the state maximum UI weekly benefits as an exogenous expansion to complete the model identification. Our identification also depends on the variation in UI benefits across states and over time. Although testing the sufficient condition of exogeneity is difficult, we can still perform certain placebo tests.

The first test, a falsification test, utilizes individual's smoking history. Based on the information on when an individual quits smoking and the smoking status of past 12 months, we construct the smoking participation status for past 12 months and the smoking cessation status for past 12 months. We define 12-month-ago smoking participation as whether an individual reports to smoke around 12 months ago; 12-mont-ago smoking cessation[23] is defined as whether an individual had quit smoking between the last 24 and 12 months (This is similar to the definition of current cessation which is defined as if one had quit smoking over the last 12 months). The underlying intuition is that the12-month-ago smoking status of individuals who lost job recently are not likely to be affected by UI benefits unless UI benefits are correlated with some unobservable variables that affect smoking behaviors. We use the same model specification in Table 4 and restrict our sample to job losers within an unemployment spell of maximum 26 weeks. We then estimate the impacts of UI benefits on smoking behaviors 12 months ago. Results in Table 5 reveal that, as expected, an increase in UI benefits has no statistically significant influence on job losers' 12-month-ago smoking participation or smoking cessation. 12-month-ago smoking participation is highly correlated with the current participation with a correlation coefficient of 0.9163. Therefore, it is not surprising to have the result in column 1 similar to estimates in Table 4. The effects of UI benefits on participation and cessation are very small, and the effect on cessation changes sign in the state-specific linear trend model.

6.5.3. Placebo test: UI-Ineligible groups

We also provide placebo tests to examine the effects of UI benefits on UI-ineligible workers, such as the employed, those not in labor force, and those who are job leavers or re-/new entrants to the labor market. The rationale is that if variations in UI benefits

[20] Kuka (2018) calculates the simulated replacement rate using the sample from the Survey of Income and Program Participation.

[21] https://oui.doleta.gov/unemploy/ui_replacement_rates.asp

[22] Individuals might not claim the benefits as soon as they become unemployed, and there might be some time delays because they have to go to the agency to submit applications.

[23] The definition of 12-month-ago smoking cessation is as follows. The variable equals 1 if the respondent was a smoker 24 months ago but did not smoke 12 months ago. The variable equals 0 if the respondent was a smoker 24 months ago and smoked 12 months ago. The mean value is 0.067, which is comparable to current cessation.

Table 6
Effect of UI on Smoking – Placebo Test on UI-Ineligibles.

VARIABLES	(1) Participation	(2)	(3) Cessation	(4)
Panel A: UI-Ineligible Typed Unemployed				
UI ($100)	−0.014	−0.007	0.009	0.019
	(0.009)	(0.016)	(0.011)	(0.018)
Observations	17,852	17,852	5,251	5,251
Panel B: Employed				
UI ($100)	0.002	0.004	0.003	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)
observations	688,706	688,706	119,779	119,779
Panel C: Not in Labor Force				
UI ($100)	−0.002	−0.005	−0.001	0.006
	(0.006)	(0.007)	(0.006)	(0.004)
Observations	138,129	138,129	29,295	29,295
Demographic Characteristics	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes

The UI are all adjusted to dollars in 2011. All samples are from 21 to 54 years old. The UI-Ineligible Typed Unemployed consists of job leavers, those with recall expectations, and new/re-entrants. The marginal effects are reported. Standard errors in the parentheses are clustered at the state level. Significance level: *** at 1%, ** at 5%, * at 10%.

are endogenous because of the omission of some unobservable state-level variables, then UI benefits are very likely to pick up the effects of unobservable state-level variables in the residual and thus have non-ignorable effects on the smoking behaviors of UI-ineligible unemployed. The estimates in Table 6 suggest that the maximum UI weekly benefits do not affect the smoking behaviors of UI-ineligible workers. Focusing on the UI-ineligible typed unemployed, we observe that the marginal effects of UI benefits are negative on smoking participation and positive on smoking cessation, even though they are not significant from zero. This is possibly because a small proportion of people in this group in reality receive UI benefits, even though they are supposed to be ineligible for UI. From Appendix Table A1, we can see that 18.7% of job leavers and 15.4% of re-entrants receive UI benefits in 2005. Because the UI receipt rate is relatively low in the UI-ineligible unemployed group, the impacts of UI benefits on smoking are not strong substantially.

6.5.4. *Triple difference model*

Our state and year FEs model is a difference-in-differences estimator, comparing changes in smoking behaviors among the unemployed workers in states where the UI increased more to states where it increased less. Kuka (2018) estimates a triple difference model in which employed workers are added as a control group for the unemployed. We use a similar approach in Appendix Table A4. In Panel A, we use UI-eligible-type unemployed as the treatment group and the employed as the control group. Results show that UI benefits can significantly increase smoking cessation. As a placebo test, in Panel B, we re-estimate the triple difference model using UI-ineligible-type unemployed (those with recall expectations, job leavers, and new/re-entrants) as the treatment group and the employed as the control group. As expected, we do not find any significant results.

6.6. *Sample selection: does UI change the composition of the unemployed?*

In this section, we address one major criticism—potential sample selection bias—for the preceding analysis because we consider only the currently unemployed sample. The literature provides strong evidence that unemployment duration and layoff possibility of individuals are functions of the UI (Feldstein, 1978; Topel, 1983; Meyer, 1989). High UI benefits reduce the opportunity costs of job loss; thus, some marginally unemployed workers are willing to quit jobs and search for more favorable jobs. As asserted by Gruber (1997), the potential problem in only focusing on the unemployed is that the marginal individuals, who experience unemployment, are systematically different from the average unemployed workers. For example, the marginally unemployed induced by high UI benefits experience fewer unemployment shocks, resulting in less smoking. This systemic difference might lead to overestimation of smoking-prevention effects.

Our sample comprises only job losers who have lost their job involuntarily; we believe most of them are not the marginally unemployed. Excluding job leavers who possibly quit jobs because of high UI benefits from the sample would help us mitigate the overestimation problem. However, in a state with generous UI, some people may be found who might be lazy to work and get fired to be job losers. In the preceding analysis, we cannot eliminate this type of the moral hazard issue. Nevertheless, we generate three outcomes that indicate whether an individual is unemployed or not to assess the potential sample selection bias in our estimates.

In columns 1 and 2 of Table 7, we define the unemployment status as 1 if the individual is unemployed and 0 if he or she is employed. The estimates show that the increase in the UI benefit level does not significantly result in unemployment. Considering that our sample only includes job losers, we re-define the unemployment status as 1 if an individual is a job loser and 0 if employed. The estimates in columns 5 and 6 show that the UI benefit level imposes no significant effects on the newly defined unemployment outcome. In columns 7 and 8, we estimate the impacts of UI benefits on the proportion of job losers in the unemployed. We define a job loser as 1 if an individual is a job loser and 0 if he or she is either a job leaver, a re/new-entrant, or a person with recall expectation. The estimates are still not statistically significant. These results suggest that the sample selection bias is less severe in our analysis.

We are also concerned that UI benefits substantially affect the unemployment duration in our data. In Appendix Table A5, we conduct a test focusing on two unemployment duration variables: the probability of being unemployed for less than 26 weeks and the length of the unemployment spell. In Panel A, we concentrate on the

Table 7
Effect of UI on Unemployment Decisions – Sample Selection Test.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	1: Unemployed		1: Unemployed (excluding re/new entrants)		1: job losers					
VARIABLES	0: Employed		0: Employed		0: Employed		0: Other Unemployed Types		0: Other Unemployed Types	
UI ($100)	−0.001	−0.001	−0.000	−0.000	−0.000	−0.000	0.005	0.000	−0.005	−0.005
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.012)	(0.018)	(0.011)	(0.019)
Observations	725,910	725,910	715,204	715,204	708,058	708,058	37,204	37,204	26,498	26,498
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes		Yes		Yes		Yes

The UI are all adjusted to dollars in 2011.
The unemployed is classified into 4 categories: the temporary layoffs with recall expectations, the job losers, the job leavers, and the new/re-entrants into the labor market. All samples are from 21 to 54 years old.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

total unemployed, and in Panels B and C, we separately consider the UI-ineligible-type unemployed (excluding re/new entrants) and UI-eligible type unemployed. The estimates in all panels show that in our smoking data, the unemployment duration is not affected by UI benefits.

The changing UI benefits that affect the composition of the unemployed should also be possibly addressed. To address this concern, we conduct a series of balance test, shown in Appendix Table A6a and Table A6b. We regress individuals' demographic characteristics on UI benefits among several different samples. These characteristics include age, gender, race, education, marriage, and number of children. The results indicate that the individuals' characteristics are not substantially correlated with UI benefits.

7. Discussion

We discuss what channels help UI benefits to prevent the smoking behavior of the unemployed. One of the most important mechanisms is the income effect. A substantial number of studies have explored the income elasticity of smoking. For example, Colman and Remler (2008) and Cheng and Kenkel (2010) find that in the US, the income effect is negative for cigarettes. The estimated income elasticity of smoking participation varies from −0.18 to −0.286. This indicates that cigarettes are inferior goods. However, Peck (2011) indicates that cigarettes are normal goods in developing countries. Our result in column 5 of Table 4 shows that a 100% increase in the weekly UI benefit level reduces smoking participation by 6.3 percentage points (average UI is $393.87, $0.016 \times 393.87/100 = 0.063$), which is 17.1% (average participation is 0.368, 0.063/0.368 = 17.1%). If the UI benefit is the only source of income for the unemployed, then the implied income elasticity of smoking participation is −0.171 if the UI benefit taking-up rate is 100%. Appendix Table A1 suggests the taking-up rate is 35.6% among all job losers. However, this rate should be higher in our sample because our sample is restricted to job losers whose unemployment duration is less than 26 weeks. Assuming the taking-up rate is 50%, the implied income elasticity of smoking participation is −0.342[24]. This is slightly higher than prior estimates but still consistent with previous studies conducted in the United States.

The other important channel that UI can also affect the smoking behaviors of the unemployed is the psychological effect. Unemployed people are more likely to smoke than employed people because the unemployed suffer from stress associated with unemployment shock. When UI benefits increase, it can improve the financial status and relieve the stress, consequently leading to reduced smoking behaviors of the unemployed. Studies such as those of Cylus et al., (2014),; Cylus et al. (2015); Gruber (1997); Kuka (2018), and Walter et al. (2014) have suggested that increasing UI benefits helps to mitigate the financial burden and reduces the stress of unemployment. Tefft (2011) uses the data on web searches for "depression" and "anxiety" from the Google Insight for Search to measure the mental health status of the general population in that state and reveals a negative association between state UI claims and the online depression-related index searches in states. This study provides some evidence that the coverage of UI benefits helps reduce the pressure level.

The rest of the section concentrates on the discussion on the psychological effect from the perspective of alleviation of the financial

[24] Column 10 of Table 4 shows that the marginal effect on cessation is 0.029, which means a 100% increase in UI can enhance cessation by 11.4 percentage points. Since our sample is adults, the change in participation is mainly driven by cessation rather than initiation. The increase in cessation by 11.4 percentage points is the same as a decrease in participation by 4.2 percentage points or 11.4%. The implied income elasticity of smoking participation is -0.228 for a taking-up rate of 50%.

Table 8
Effect of UI on Spouse's Labor Market Outcomes.

VARIABLES	(1) Work or Not	(2)	(3) Work Hours if Work	(4)	(5) Overwork if Work	(6)
Panel A: Spouses of UI-Eligible Sample						
UI ($100)	0.005	−0.001	−0.676*	−0.468	−0.034**	−0.018
	(0.022)	(0.038)	(0.391)	(0.602)	(0.017)	(0.022)
Observations	6,938	6,938	4,749	4,749	4,749	4,749
Panel B: Spouses of UI-Ineligible Typed Unemployed (excluding those with recall expectations)						
UI ($100)	0.019	0.025	−0.436	−0.377	−0.012	−0.013
	(0.018)	(0.028)	(0.440)	(0.614)	(0.016)	(0.017)
Observations	6,173	6,173	4,438	4,438	4,438	4,438
Panel C: Spouses of Employed						
UI ($100)	0.000	−0.000	−0.068	−0.102*	−0.004*	−0.003
	(0.002)	(0.002)	(0.069)	(0.059)	(0.002)	(0.002)
Observations	496,445	496,445	394,868	394,868	394,868	394,868
Panel D: Spouses of those Not in Labor Force						
UI ($100)	−0.001	−0.007	−0.016	−0.068	0.001	0.003
	(0.006)	(0.009)	(0.150)	(0.161)	(0.004)	(0.005)
Observations	96,582	96,582	76,744	76,744	76,744	76,744
Demographics	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes		Yes

The UI are all adjusted to dollars in 2011.
The UI-Eligible Sample includes spouse of those unemployed who are from 21 to 54, lose their job involuntarily and are in an unemployment spell of less than 26 weeks.
UI-ineligible Unemployed includes job leavers and re/new-entrants in the market.
The overwork outcome is defined by a threshold of 40 (1–39 hours/week: regular work, 40+/weeks: overwork).
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

pressure. The CPS-TUS data set, however, does not provide us with the detailed information of the mental health status; therefore, we cannot directly investigate the effects of increasing UI benefits on the mental health status. It is a less than ideal method to test the psychological effect from some indirect aspects. In the following sections, we describe some suggestive evidence.

First, the income effect and psychological effect take place only if there is a positive relationship between UI benefits in law and actual unemployment benefits received. We explore this by regressing the state actual weekly UI benefits on the state maximum UI benefit in law in Appendix Table A7. The result shows that a $100 increase in the maximum weekly UI benefits in law results in at least $34.1 increase in actual weekly UI benefits.

7.1. Suggestive Evidence-I: impacts of unemployment insurance on spouse's working hours

If an individual falls into an unexpected unemployment, his or her spouse is likely to work longer hours to compensate the family income loss due to the job loss. We test how increasing UI benefits affect the employed spouses' labor force decisions after individuals lose their jobs involuntarily. We define three labor market outcomes: working or not, the weekly working hours if working, and the over-work status (a binary variable, which has a value of 1 if working hours are at least 40 per week and 0 if the working hours are positive but less than 40 per week).

Table 8 reports the estimated impacts of UI benefits on spouses' working status. In Panel A, we focus on spouses of the UI-eligible unemployed (unemployment spell of less than 26 weeks). We find that UI benefit level does not affect whether the spouse works or not. However, we find some suggestive evidence that UI can affect the working hours if the spouse works. This impact is very small. When the UI benefit level increases by $100, the spouses' over-work possibility decreases by 3.4 percentage points in column 5[25]. We also provide a series of placebo tests using spouses of different UI-ineligible samples in Panels B, C, and D. In general, we do not find strong impacts of UI benefits on the labor outcomes of the spouses of neither the employed, those not in the labor force, or the UI-ineligible unemployed. In addition, we explore the effects of UI on spouses' smoking behaviors. We find that the increase in the weekly UI benefits can reduce smoking participation of the spouse[26]. Among other UI-ineligible groups, the spouses' smoking behaviors are not responsive to the UI benefit expansion. This evidence sheds some light on the fact that UI benefits might mitigate the financial pressure and burden; thus, household members do not need to work long time to compensate the loss of income due to job loss.

7.2. Suggestive Evidence-II: heterogeneous impacts of unemployment insurance

In Table 9, we compare the impacts of UI benefits on smoking participation in groups with different potential financial pressure levels. Limited to the cessation sample size, we do not observe any precise estimates for cessation.

First, we check in a downward economy, given the unemployed have fewer chances to get back to jobs and bear more severe financial burden, whether a generous UI program would be more helpful. In columns 1 and 2, in addition to control for UI benefits and state

[25] Given the mean of the spouses' over-work possibility is 78.08%, a $100 increase in the UI benefits decreases the probability of over-working by 4.4% (=3.4/78.08). However, the results are not robust to the inclusion of state-specific time trends.

[26] The estimated marginal effect is -0.048 and statistically significant at the conventional level. The magnitude appears to be extremely large. Results are available upon request.

Table 9
Effect of UI on Smoking Participation – Heterogeneous Effects.

VARIABLES	(1) By State Unemployment Rate	(2)	(3) By Education	(4)	(5) By Age	(6)
UI ($100)	0.019	−0.030	−0.131***	−0.114**	−0.077	−0.059
	(0.070)	(0.071)	(0.046)	(0.054)	(0.048)	(0.057)
UI ($100) * State Unemployment Rate	−0.012	−0.003				
	(0.009)	(0.010)				
UI ($100) * More than High School			0.087**	0.088**		
			(0.041)	(0.042)		
UI ($100) * Age					0.000	0.000
					(0.001)	(0.001)
Observations	13,800	13,800	13,800	13,800	13,800	13,800
Demographic Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes		Yes

The UI are all adjusted to dollars in 2011.
The UI-Eligible Sample includes people who are from 21 to 54, lose their job involuntarily and are in an unemployment spell of less than 26 weeks.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

unemployment rate, we add the interaction of UI benefits and the state unemployment rate into our model. The estimates illustrate that in states with a higher unemployment rate, if UI benefits increase by $100, the smoking participation of the unemployed reduces by 1.2 percentage points more compared with that in states with a lower unemployment rate. However, it is not significant from zero.

Secondly, individuals with high human capital are more competitive and popular in the labor market compared with their counterparts. If highly educated people get laid off, they have more access to get re-employed, which mitigates their economic pressure. In columns 3 and 4, we add the interaction of UI benefits and the binary variable indicating whether the person has a high school degree or more. Marginal effect of this interaction term is positive implying that the impacts of UI benefits among highly educated individuals are less strong. That is, people with lower social status (i.e., lower education level) and less competitive in the labor market benefit more from generous UI benefits. We add the interaction term of UI benefits and the age into the model in columns 5 and 6. The estimation results show that there is no significant difference in the impacts of UI benefits across age.

In summary, we estimate the health effects of UI benefits in different groups and reveal that individuals with severe financial pressure tend to be more responsive to the increasing UI benefits. However, we have to be cautious in this interpretation. As shown in Appendix Table A1, the UI benefit taking-up rate is relatively low. In different states and time periods, the taking-up rate could also be different. Therefore, the differential taking-up rate can also be one factor that drives our estimates in Table 9. If the purpose of the UI policy is to help those facing difficult economic conditions after losing jobs, then our results illustrate that the state UI benefits generate some positive health effects, at least to some extent.

8. Conclusion

This study expands previous research by substantiating the link between social assistance policies and individuals' health behaviors. Studies such as those of Cylus et al. (2015) and Tefft (2011) have focused on the causal relationship between UI benefits and health status. Understanding how individuals' health behaviors are affected by UI benefits can reveal some of the mechanisms that explain the association between social assistance program and health status.

We also expand the previous studies, such as those of Gruber (1997) and Kuka (2018), by distinguishing a UI-eligible sample from the entire unemployed population. Kuka (2018) uses the entire unemployed sample and identifies no significant effect of UI benefits on people's health-risk behaviors; our results show that a large number of non-UI target subgroups cover the real effects.

We focus on the UI-eligible job losers and reveal that a higher UI benefit program can significantly increase smoking cessation. Specifically, a $100 increase in the maximum UI weekly benefits can enhance the smoking cessation by approximately 2.9 percentage points.

Our suggestive evidence suggests that the spouses of the UI-eligible unemployed workers may have a lower possibility of overwork in a higher UI benefit system. In addition, compared with the more educated unemployed group, the less educated group who is less competitive in the labor market is more responsive to the UI benefits. These results support our assumption that UI benefits prevent smoking by easing financial pressure. Finally, our results are robust to various sensitivity checks.

Furthermore, we need to be cautious about the interpretation of our results. We cannot eliminate that the possibility that the differential taking-up rate can result in the heterogeneous effect of our estimates in different subgroups. In addition, our results are not robust to the inclusion of the sampling weights. This is largely because we set several constraints to our sample. Thus, we should be cautious about the representation of our sample when we interpret our results.

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AppendixA.

Table A1
Statistics of the UI Application and Taking-up Rate.

	Women	Men	Total
Panel A: UI Application Rates among all Unemployed Workers			
Job losers	50.1	51	50.7
Job leavers	21.1	16.2	18.7
Re-entrants	18.1	12.1	15.4
All unemployed	33.5	35.9	34.8
Panel B: UI Taking-up Rates among all Unemployed Workers			
Job losers	37	34.8	35.6
Job leavers	10.1	7.4	8.8
Re-entrants	12.3	9	10.9
All unemployed	23.6	24.3	23.9
Data Sources: Vroman (2009).			

Table A2
Effect of UI on Smoking – UI Measurement Sensitivity Test.

VARIABLES	(1) Participation	(2)	(3) Cessation	(4)
Panel A: Using Log of UI				
log UI	−0.090*	−0.065	0.120**	0.127**
	(0.046)	(0.061)	(0.052)	−0.052
Observations	13,800	13,800	4,177	4,177
Panel B: UI Replacement Ratio-1 = wgt. ave. of (WBA)/(hourly wage * 40 hrs)				
UI Replacement Ratio-1	−0.110	0.202	0.345*	0.469**
	(0.280)	(0.330)	(0.199)	(0.232)
Observations	11,549	11,549	3,379	3,379
Panel C: UI Replacement Ratio-2 = ratio of (wgt. ave. WBA)/(wgt. ave. normal hourly wage * 40 hrs)				
UI Replacement Ratio-2	−0.115	0.138	0.455***	0.515**
	(0.223)	(0.265)	(0.176)	(0.218)
Observations	11,549	11,549	3,379	3,379
Demographic Characteristics	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes

The UI are all adjusted to dollars in 2011.
The UI-Eligible Sample includes people who are from 21 to 54, lose their job involuntarily and are in an unemployment spell of less than 26 weeks. The two UI replacement ratios are obtained from the Department of Labor. We lose some observations in Panel B and C because the UI replacement ratios can only be tracked back to 1997. Thus we lose the observations in 1995 and 1996.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

Table A3
Effect of UI on Smoking – Robustness Test to Sample Specification.

VARIABLES	(1) Participation	(2)	(3) Cessation	(4)
Panel A: Involuntary Job losers, Unempl. Dur to 30 weeks				
UI ($100)	−0.021*	−0.015	0.025*	0.028**
	(0.011)	(0.015)	(0.013)	(0.014)
Observations	14,413	14,413	4,344	4,344

Table A3 (*Continued*)

VARIABLES	(1) Participation	(2)	(3) Cessation	(4)
Panel B: Involuntary Job losers, Unempl. Dur to 52 weeks				
UI ($100)	−0.020**	−0.010	0.022*	0.033**
	(0.010)	(0.014)	(0.013)	−0.015
Observations	15,949	15,949	4,731	4,731
Panel C: Involuntary Job losers + With Recall Expectations, Unempl. Dur. < = 26 weeks				
UI ($100)	−0.022*	−0.027**	0.020	0.018
	(0.012)	(0.014)	(0.013)	(0.015)
Observations	16,326	16,326	4,933	4,933
Demographic Characteristics	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes

The UI are all adjusted to dollars in 2011.
All samples are from 21 to 54 years old.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

Table A4
: Effect of UI on Smoking – Triple Difference.

VARIABLES	(1) Participation	(2)	(3) Cessation	(4)
Panel A: UI-Eligible Typed Unemployed (Treated) vs. Employed (Control)				
UI-Eligible Typed Unemployed * UI ($100)	−0.008*	−0.008*	0.009**	0.010**
	(0.004)	(0.004)	(0.005)	(0.005)
UI ($100)	0.002	0.005	0.003	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)
UI-Eligible Typed Unemployed	0.124***	0.123***	−0.057***	−0.057***
	(0.018)	(0.018)	(0.020)	(0.021)
Observations	708,058	708,058	125,554	125,554
Panel B: UI-Ineligible Typed Unemployed (Treated) vs. Employed (Control)				
UI-Ineligible Typed Unemployed * UI ($100)	−0.005	−0.005	−0.002	−0.002
	(0.004)	(0.004)	(0.004)	(0.004)
UI ($100)	0.001	0.004	0.003	−0.002
	(0.003)	(0.003)	(0.003)	(0.003)
UI-Ineligible Typed Unemployed	0.103***	0.102***	0.003	0.002
	(0.018)	(0.018)	(0.019)	(0.019)
Observations	706,558	706,558	125,030	125,030
Demographic Characteristics	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes

The UI are all adjusted to dollars in 2011.
The UI-Eligible Typed Unemployed consists of all the job losers, and the UI-Ineligible Typed Unemployed consists of those with recall expectations, leavers and the new/re-entrants into the labor market. All individuals are from 21 to 54 years old.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

Table A5
: Effect of UI on Unemployment Durations.

VARIABLES	(1) 0: Being Unemployed >26 weeks / 1: Being Unemployed <=26 weeks	(2)	(3) Unemployment weeks	(4)
Panel A: Total Unemployed Sample				
UI ($100)	−0.004	0.007	−0.160	−0.247
	(0.011)	(0.015)	(0.532)	(0.579)
Observations	37,204	37,204	37,204	37,204
Panel B: Unemployed Sample excluding re/new entrants				
UI ($100)	0.005	0.005	−0.420	−0.027
	(0.010)	(0.015)	(0.538)	(0.731)
Observations	26,498	26,498	26,498	26,498
Panel C: Job losers				
UI ($100)	−0.001	0.008	−0.184	−0.174
	(0.012)	(0.016)	(0.630)	(0.812)
Observations	19,352	19,352	19,352	19,352
Demographics	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes

The UI are all adjusted to dollars in 2011. All samples are from 21 to 54 years old.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

Table A6a
: Effect of UI on Demographics – Balance Test.

VARIABLES	(1) Age	(2)	(3) Female	(4)	(5) White	(6)	(7) Less Than High School	(8)
Panel A: Total Unemployed Sample								
UI ($100)	−0.025	0.127	−0.005	−0.007	−0.022**	−0.014	−0.000	−0.003
	(0.184)	(0.178)	(0.007)	(0.011)	(0.009)	(0.013)	(0.007)	(0.011)
Observations	37,204	37,204	37,204	37,204	37,204	37,204	37,204	37,204
Panel B: Unemployed Sample excluding re/new entrants								
UI ($100)	0.006	0.382	−0.002	−0.011	−0.017**	−0.002	−0.002	−0.005
	(0.222)	(0.245)	(0.011)	(0.014)	(0.008)	(0.015)	(0.008)	(0.010)
Observations	26,498	26,498	26,498	26,498	26,498	26,498	26,498	26,498
Panel C: Job losers								
UI ($100)	−0.074	0.426	−0.003	−0.025	−0.018*	−0.000	−0.006	−0.013
	(0.285)	(0.256)	(0.015)	(0.022)	(0.011)	(0.019)	(0.011)	(0.017)
Observations	19,352	19,352	19,352	19,352	19,352	19,352	19,352	19,352
Panel D: UI-Eligible Sample (Job losers & unemployed duration <=26 weeks)								
UI ($100)	0.046	0.433	−0.010	−0.028	−0.015	−0.004	−0.005	−0.012
	(0.278)	(0.313)	(0.015)	(0.022)	(0.011)	(0.016)	(0.009)	(0.013)
Observations	13,800	13,800	13,800	13,800	13,800	13,800	13,800	13,800
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes		Yes		Yes

The UI are all adjusted to dollars in 2011. All samples are from 21 to 54 years old.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

Table A6b
: Effect of UI on Demographics – Balance Test (Continued).

VARIABLES	(1) High School	(2)	(3) More Than High School	(4)	(5) Married	(6)	(7) Number of Children	(8)
Panel A: Total Unemployed Sample								
UI ($100)	−0.005	−0.004	0.007	−0.001	−0.011**	−0.007	0.018	0.008
	(0.009)	(0.011)	(0.008)	(0.011)	(0.006)	(0.008)	(0.016)	(0.018)
Observations	37,204	37,204	37,204	37,204	37,204	37,204	37,204	37,204
Panel B: Unemployed Sample excluding Not in Labor Force								
UI ($100)	−0.004	0.002	0.003	−0.014	−0.005	−0.004	−0.000	−0.011
	(0.012)	(0.013)	(0.008)	(0.012)	(0.006)	(0.012)	(0.016)	(0.021)
Observations	26,498	26,498	26,498	26,498	26,498	26,498	26,498	26,498
Panel C: Job losers								
UI ($100)	−0.006	0.006	0.012	−0.004	0.002	0.012	0.003	−0.024
	(0.013)	(0.016)	(0.009)	(0.014)	(0.008)	(0.010)	(0.019)	(0.024)
Observations	19,352	19,352	19,352	19,352	19,352	19,352	19,352	19,352
Panel D: UI-Eligible Sample (Job losers & unemployed duration <=26 weeks)								
UI ($100)	−0.007	−0.005	0.004	0.000	0.011	0.016	−0.017	−0.023
	(0.013)	(0.019)	(0.008)	(0.012)	(0.008)	(0.013)	(0.025)	(0.031)
Observations	13,800	13,800	13,800	13,800	13,800	13,800	13,800	13,800
Demographics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other Welfare Policies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Smoking Policies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend		Yes		Yes		Yes		Yes

The UI are all adjusted to dollars in 1992. All samples are from 21 to 54 years old.
The marginal effects are reported. Standard errors in the parentheses are clustered at the state level.
Significance level: *** at 1%, ** at 5%, * at 10%.

Table A7
Effect of Maximum UI Benefit on Actual UI Benefit.

VARIABLES	(1) Dependent Variable: State Actual Weekly UI Benefit ($100)	(2)	(3)	(4)	(5)
UI ($100)	0.364***	0.357***	0.348***	0.353***	0.341***
	(0.046)	(0.046)	(0.047)	(0.048)	(0.067)
Constant	−4.260***	−3.553***	−5.394***	−5.850***	−5.666***
	(1.399)	(1.352)	(1.619)	(1.551)	(2.180)
Observations	550	550	550	550	550
R-squared	0.954	0.955	0.957	0.958	0.972
State Characteristics	Yes	Yes	Yes	Yes	Yes
UI maximum weeks length		Yes	Yes	Yes	Yes
Other Welfare Policies			Yes	Yes	Yes
Smoking Policies				Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
State-Specific Linear Trend					Yes

The UI are all adjusted to dollars in 2011.
We only use the state-year pairs that appear in our baseline.
Robust standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

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